

A Very Short Introduction to OTFS – Orthogonal Time Frequency Spacing

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ABSTRACT

This 5-page file, serving as a course report for IERG5110 (fall 2024), aims to give a simple introduction to the concept of OTFS. Our main references are the first four chapters of [1].

Index Terms— Orthogonal Time Frequency Spacing (OTFS); Delay-Doppler Communication;

1. REPRESENTATION OF WIRELESS CHANNELS

An inevitable concept when introducing OTFS is the “delay-Doppler” domain modulation. Although the study of “delay-Doppler” domain in fact is a rather old topic in radar theory (see e.g., [2]), its involvement in communication is quite recent, given OTFS was proposed by Hadani et al. in [3] in 2017.

It is good to know that delay and Doppler (Doppler shift) are nothing but just some offset or shift in time and frequency domain, respectively. To be precise, suppose some signal $s(t)$ should be f_c Hz, but is eventually received as $f_c + f_d$ Hz, then the additional amount f_d is termed as “Doppler shift”, sometimes abbreviated as “Doppler”. With this, note that a Doppler shift by f_d Hz is equivalent to modulating the transmitted signal $s(t)$ with $e^{j2\pi f_d t}$.

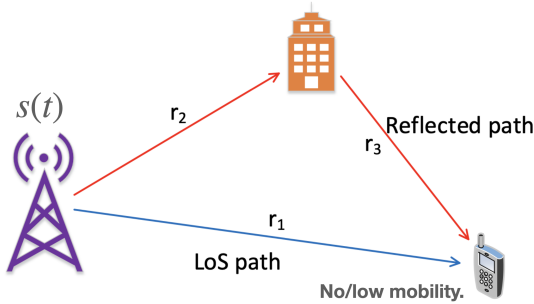


Fig. 1: A static multi-path channel with $P = 2$ paths. Copied and modified from Fig. 2.1 of [1]

Now consider a (low-mobility) wireless channel according to fig. 1. At the transmitter, a (baseband) signal $s(t)$ is transmitted. One can easily derive that the received signal as

$$r(t) = g_1 s(t - \tau_1) + g_2 s(t - \tau_2), \quad (1)$$

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where g_1, g_2 denotes the complex channel gain of each path. $\tau_1 = \frac{r_1}{c}$ and $\tau_2 = \frac{r_2 + r_3}{c}$ are the delays of the two paths, and $c = 3 \times 10^8$ m/s denotes the speed of light. In this case, $\tau_2 - \tau_1$ is the delay spread. In general, delay spread means the time difference between the last arrived path and the first arrived path.

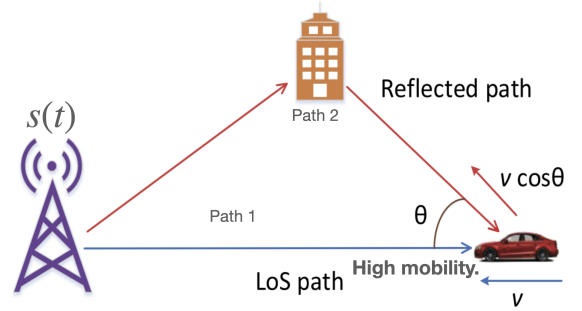


Fig. 2: A high-mobility multi-path channel with $P = 2$ paths. Copied and modified from Fig. 2.2 of [1]

The needs for communication in a non-static scenario have emerged, due to the development of high-speed means of transport, e.g., high-speed trains. This pushes people to step out of the comfort zone of static channel, and consider channels with mobility.

In the following, we consider a “high-mobility” channel according to fig. 2. Not hard to derive the Doppler shifts for path $i, i = 1, 2$ are $\nu_1 = \frac{v}{c} f_c$ and $\nu_2 = \frac{v \cos \theta}{c} f_c$. So the received signal is:

$$r(t) = g_1 e^{j2\pi \nu_1 (t - \tau_1)} s(t - \tau_1) + g_2 e^{j2\pi \nu_2 (t - \tau_2)} s(t - \tau_2). \quad (2)$$

Note that if there is no mobility, then those $\nu_i = 0$, and the model degenerates back to the static channel model in eq. (1).

One may inclined to think that $\nu \approx \frac{v}{c} f_c$ is a negligible quantity, since $v \ll c$. However, f_c is usually a large quantity, which in turn makes ν not too small. For example, when $f_c = 2$ GHz (WiFi level), $v = 150$ m/s (twice of a usual car speed), then $\nu = \frac{v}{c} f_c = 1$ KHz.

In general, we can consider a P -path scenario as the fol-

lowing, by generalizing eq. (2):

$$r(t) = \sum_{i=1}^P g_i e^{j2\pi\nu_i(t-\tau_i)} s(t-\tau_i) \quad (3)$$

$$= \sum_{i=1}^P g_i e^{j2\pi\nu_i t} s(t-\tau_i) e^{-j2\pi\nu_i \tau_i}. \quad (4)$$

If one defines the delay-Doppler channel representation as

$$h(\tau, \nu) = \sum_{i=1}^P g_i e^{-j2\pi\nu_i \tau_i} \delta(\tau - \tau_i) \delta(\nu - \nu_i), \quad (5)$$

then the received signal $r(t)$ can be written as (not hard to see):

$$r(t) = \iint h(\tau, \nu) e^{j2\pi\nu t} s(t-\tau) d\nu d\tau. \quad (6)$$

Note that eq. (5) induces some sort of sparsity, since it can be completely specified by these quantities: $(g_i, \nu_i, \tau_i), i \in [P]$. In the following, we will relate eq. (5) with other channel models that we are already more familiar with.

Specifically, we will cover delay-time representation and frequency-time representation.

1.1. Delay-time representation

If we go back to eq. (3), and define $g(\tau, t) \triangleq g_i e^{j2\pi\nu_i(t-\tau)} \delta(t-\tau)$, then one can rewrite eq. (3) as

$$r(t) = \int_0^\infty g(\tau, t) s(t-\tau) d\tau = \sum_{i=1}^P g(\tau_i, t) s(t-\tau_i). \quad (7)$$

1.2. Frequency-time representation

Take eq. (7), and denote the Fourier transform of r as R . We see that

$$r = g_t * s \iff R(f) = H_t(f) \cdot S(f) = H(f, t) \cdot S(f). \quad (8)$$

Then we see that

$$r(t) = \mathcal{F}^{-1}(R(f)) = \int H(f, t) S(f) e^{j2\pi f t} df, \quad (9)$$

where $\mathcal{F}^{-1}(\cdot)$ denotes the inverse Fourier transform. We call this $H(f, t)$ as the frequency-time domain channel representation.

Remark 1. Example, and schemes of communication constructed based on these two channel representations exist in the literature, due to their intuitiveness. The advantage of the DD representation is its sparsity, which can be demonstrated via these several fig. 3, fig. 4 and eq. (5). I have to admit that these are not characterizing the same channel, but gathered from different sources. The mere purpose here is to show that DD representation is sparse, comparing with the other two representations.

• $g(\tau, t)$ be like :(

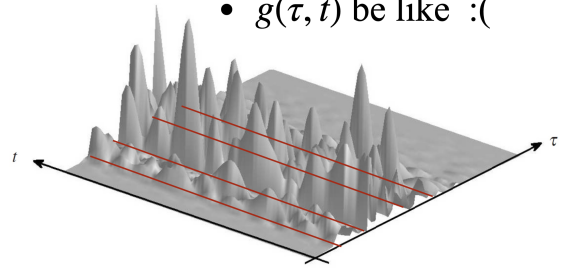
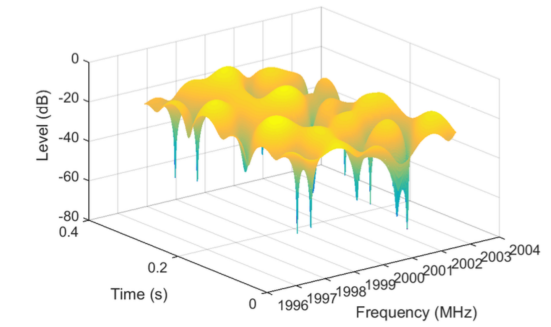


Fig. 3: Time variant impulse response of some $g(\tau, t)$.



• $H(t, f)$ be like :(

Fig. 4: Frequency-time response of some $H(f, t)$.

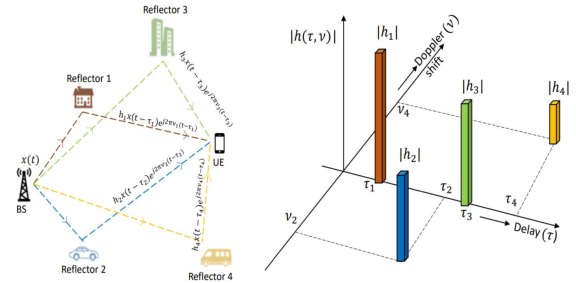


Fig. 5: DD domain response of some $h(\tau, \nu)$ (in absolute value) of some 4-path channel with mobility.

2. OFDM IN HIGH-MOBILITY CHANNELS

In this section, let us briefly talk about why OFDM is not perfect in high-mobility scenarios, and hence validates the necessity of OTFS, and delay-Doppler modulations.

Consider an OFDM system conducts M -point FFT, with cyclic prefix length being $l_{\max}$. Denote $x \in \mathbb{C}^M$ the information of OFDM (in frequency domain), and $\tilde{x} \triangleq \mathbf{F}_M^\dagger x \in \mathbb{C}^M$. And assume the channel is static, aka, $h(t) = \sum_{i=1}^P g_i \delta(t - \tau_i)$. Then the discrete-time equivalent channel vector can be

summarized as $h' \triangleq [h_0, \dots, h_{l_{\max}}]^T$, with $h_l = g_i$ if g_i corresponds to the l -th delay tap, $i \in [P]$. Then the baseband equivalent signal (including cyclic prefix) of OFDM can be given by

$$r' = h' * s' + w' \in \mathbb{C}^{M+l_{\max}}. \quad (10)$$

After taking away the CP, we have

$$r = r'[l_{\max} : l_{\max} + M - 1] = h \otimes \tilde{x} + w, \quad (11)$$

where $h = [h_0, \dots, h_{l_{\max}}, 0, \dots, 0]^T \in \mathbb{C}^M$. Note that the operation becomes from linear convolution to circulant convolution, which further gives rise to the following expression of r :

$$r = \mathbf{H}\tilde{x} + w, \quad (12)$$

where

$$\mathbf{H} = \begin{bmatrix} h_0 & 0 & \dots & 0 & h_{l_{\max}} & h_{l_{\max}-1} & \dots & h_1 \\ h_1 & h_0 & \dots & 0 & 0 & h_{l_{\max}} & \dots & h_2 \\ \vdots & \ddots & \dots & \dots & \dots & \dots & \ddots & \vdots \\ 0 & 0 & \dots & h_{l_{\max}} & h_{l_{\max}-1} & \dots & h_1 & h_0 \end{bmatrix}. \quad (13)$$

Note that $\mathbf{H} \in \mathbb{C}^{M \times M}$ is a circulant matrix. Hence can be easily diagonalized via FFT/IFFT matrices, this enables one to use a single-tap equalizer to do recovery. This nice property comes from the crucial assumption that the channel is static at the first place.

It is convenient to emphasis here that $\mathbf{H}$ has another expression:

$$\mathbf{H} = \sum_{i=1}^P g_i \mathbf{\Pi}^{l_i}, \quad (14)$$

where $\mathbf{\Pi}$ denotes the "forward shift" matrix, given by

$$\mathbf{\Pi} \triangleq \begin{bmatrix} 0 & \dots & 0 & 1 \\ 1 & \ddots & 0 & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & 1 & 0 \end{bmatrix}. \quad (15)$$

Then, if we take Doppler into consideration, this matrix $\mathbf{H}$ will become

$$\mathbf{H} = \sum_{i=1}^P g_i \underbrace{\mathbf{\Pi}^{l_i}}_{\text{time domain delay}} \underbrace{\mathbf{D}^{k_i}}_{\text{Doppler shift}}, \quad (16)$$

where $\mathbf{D} = \text{diag}(1, \omega, \dots, \omega^{M-1})$, where $\omega = e^{j2\pi/M}$ denotes the minimal unit of possible Doppler shift of our system. And such matrix $\mathbf{H}$ will no longer be circulant, hence cannot be diagonalized that easily using FFT/IFFT matrices, and hence brings some inter-carrier interference (ICI). Thus, some more sophisticated schemes than OFDM are required to cope with this.

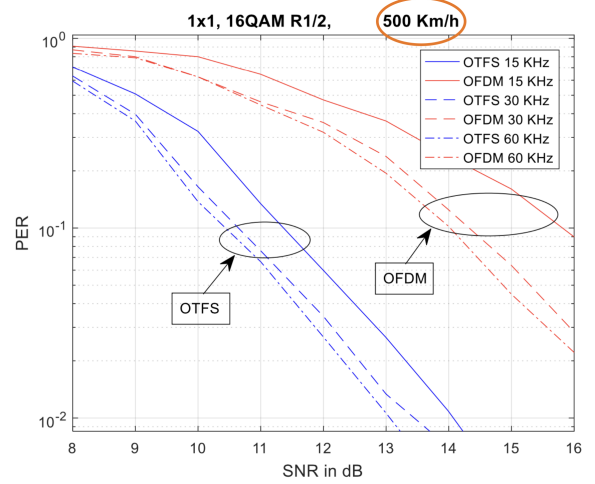


Fig. 6: Comparison between OTFS and OFDM.

Fig. 6 (high-mobility SISO channel, with a half-rate 16QAM under different f_c values) shows that OTFS indeed outperforms OFDM when Doppler is not negligible.

3. HOW DOES OTFS WORK?

In OFDM, we store our information in frequency domain vectors. In OTFS, we place them in "delay-Doppler" domain matrix $\mathbf{X} \in \mathbb{C}^{M \times N}$, with entries $\mathbf{X}[m, n]$, $m \in [M]$, $n \in [N]$. Then we first convert $\mathbf{X}$ into frequency-time domain, called $\mathbf{X}_{ft} \in \mathbb{C}^{M \times N}$ by applying (N -point) IFFT along the rows of $\mathbf{X}$, and applying (M -point) FFT along the columns of it. Put it in math, we have:

$$\mathbf{X}_{ft} = \frac{1}{\sqrt{MN}} \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} \mathbf{X}[m, n] e^{j2\pi(\frac{nk}{N} - \frac{ml}{M})} \quad (17)$$

$$= \mathbf{F}_M \underbrace{\mathbf{X}}_{M \times N} \mathbf{F}_N^\dagger. \quad (18)$$

Such operation is often referred as "inverse symplectic FFT" (ISFFT). Not surprisingly, ISFFT has a inverse mapping, called "SFFT". And these two transforms connects the frequency-time domain and DD domain. This was pictorialized in fig. 7. Then, in order to transmit, we need to further go to time domain. A frequency-time modulator converts the 2D samples $\mathbf{X}_{ft}[l, k]$ to a continuous-time waveform $s(t)$ using a transmit waveform $g_{tx}(t)$ as:

$$s(t) = \sum_{k=0}^{N-1} \sum_{l=0}^{M-1} \mathbf{X}_{ft}[l, k] g_{tx}(t - kT) e^{j2\pi l \Delta f (t - kT)}, \quad (19)$$

This operation is referred to in the literature as the "Heisenberg transform". Where Δf (space of spectrum samples) and T (duration of each block) are some parameters specified by the OTFS scheme, which are indeed crucial in practice, since

they are related to the Doppler and delay resolution of the system, but now we do not investigate deep on them.

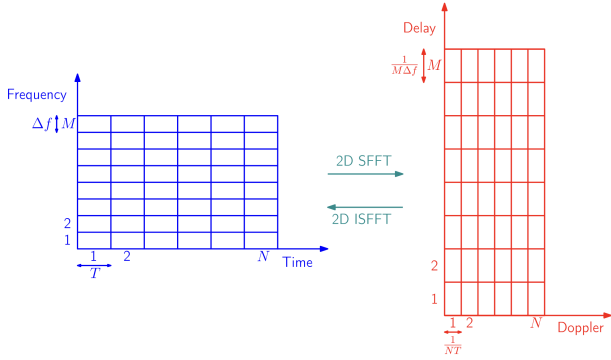


Fig. 7: Conversion between $\mathbf{X}$ and $\mathbf{X}_{ft}$. $\mathbf{X}_{ft} = \text{ISFFT}(\mathbf{X})$, $\mathbf{X} = \text{SFFT}(\mathbf{X}_{ft})$, copied from Fig. 4.2 of [1]

As we have already introduced, the channel will impair $s(t)$, leading to some

$$r(t) = \iint h(\tau, \nu) s(t - \tau) e^{j2\pi\nu(t-\tau)} d\tau d\nu \quad (20)$$

to be received.

At the receiver side, the received signal $r(t)$ is passed through a matched filter computing the cross-ambiguity function as

$$Y(f, t) = \int r(\iota) g_{rx}^*(\iota - t) e^{-j2\pi f(\iota - t)} d\iota, \quad (21)$$

and then sampling $Y(f, t)$ to form the frequency-time domain received sample matrix $\mathbf{Y}_{ft} \in \mathbb{C}^{M \times N}$, namely

$$\mathbf{Y}_{ft}[l, k] = Y(f, t)|_{f=l\Delta f, t=kT}. \quad (22)$$

This operation to construct $\mathbf{Y}_{ft}$ is often referred as "Wigner transform". Up till now, fig. 8, which is an overall diagram of OTFS modulation and demodulation procedures, should be clear.

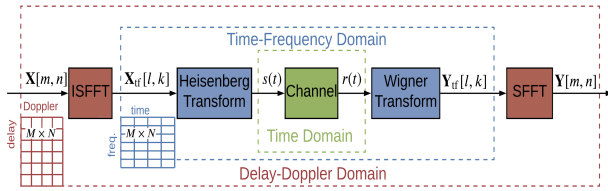


Fig. 8: OTFS system diagram (ISFFT multi-carrier construction). Copied from Fig. 4.1 of [1]

3.1. More on DD domain relations

In order to avoid the concept of "quasi-periodicity", we assume the transmit and receive side pulses g_{tx}, g_{rx} are "perfectly localized in time and frequency" (which is impossible,

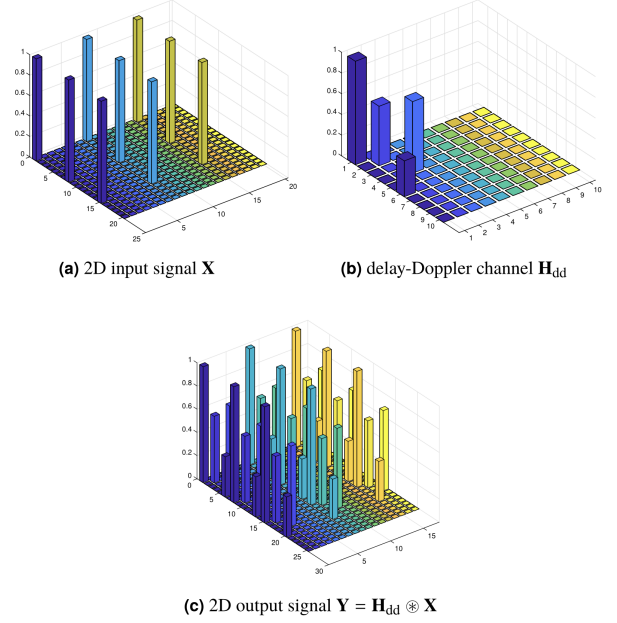


Fig. 9: Received 2D signal for ideal pulses as the 2D circular convolution of (a) and (b). Copied from Fig. 4.4 of [1]

due to the Heisenberg's uncertainty principle). With this powerful assumption, omitting the noise, we obtain

$$\mathbf{Y}_{ft}[l, k] = \mathbf{H}_{ft}[l, k] \cdot \mathbf{X}_{ft}[l, k], \quad (23)$$

where $\mathbf{H}_{ft} \in \mathbb{C}^{M \times N}$ is the frequency-time domain channel matrix, whose entries are given by:

$$\mathbf{H}_{ft}[l, k] = \iint h(\tau, \nu) e^{j2\pi\nu kT} e^{-j2\pi l \Delta f \tau} d\tau d\nu, \quad (24)$$

for $l \in [M]$ and $k \in [N]$. Note that eq. (23) is a "easier-to-write-down" equation of the input $\mathbf{X}_{ft}$ and output $\mathbf{Y}_{ft}$, and it is in frequency-time domain. If we switch everything into DD domain, namely, rewrite eq. (23) with DD domain quantities, we then have:

$$\mathbf{Y}[m, n] = \mathbf{H}_{dd}[m, n] \underset{\text{2D circular convolution}}{\circledast} \mathbf{X}[m, n] \quad (25)$$

$$\triangleq \sum_{m'} \sum_{n'} \mathbf{H}_{dd}[m', n'] \mathbf{X}[(m - m')_M, (n - n')_N] \quad (26)$$

$$= \sum_{i=1}^P g_i \mathbf{X}[(m - l_i)_M, (n - k_i)_N], \quad (27)$$

where $(a)_N \triangleq a \bmod N$.

Recall that in previous sections we have mentioned that channel representation in DD domain should be sparse, and such advantage can be better appreciated in pictorial ways, fig. 9 clearly demonstrates how the 2D circular convolution works between two matrices.

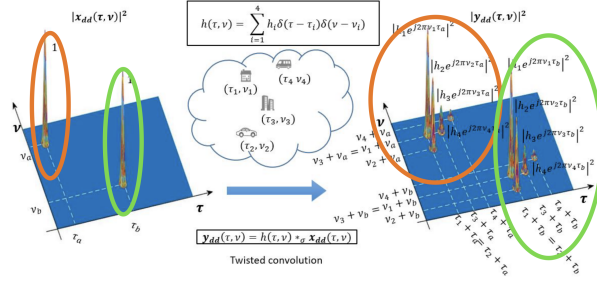


Fig. 10: 2D received signal for ideal 2-spike pulses as the 2D circular convolution with a 4-path channel. Copied and modified from Fig. 10 of [4]

4. DD DOMAIN SIGNAL PROCESSING

4.1. Channel Estimation

By fig. 9, we noticed that in the DD domain, the transmitted signal's interaction with the channel is just creating a "local replicate" of the channel impulses "beside" the transmitted signal. This can be seen in a clearer example of fig. 10. And such property may be leveraged to do channel estimation. And that was indeed proposed in the literature. Here I will only cover one method that does so in DD domain. Chapter 7 of [1] gives a way more complete introduction to OTFS channel estimation.

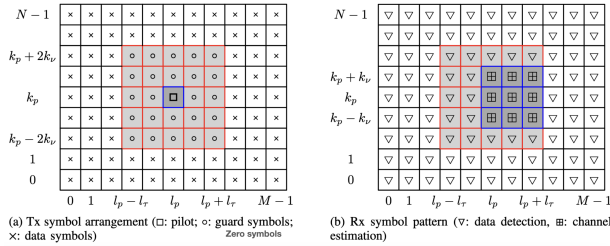


Fig. 11: Pilot arrangement of a DD domain OTFS frame. Copied from Fig. 1 in [5].

Fig. 11 demonstrate how one can add pilot to his OTFS frame. In the demonstration, the grey cells on LHS are deliberately left null (not used to transmit anything) in order to resist possible contamination from the data (white) cells by the channel. At the receiver's side, receiver uses the blue cells to do channel estimation. And then use such estimation to do data detection/decoding.

4.2. Data Recovery

Now, let us assume the channel estimation task is perfectly done. And we are head for data recovery.

Lemma 1. Let $\mathbf{Z} = \mathbf{U} \circledast \mathbf{V}$. Then $\text{vec}(\mathbf{Z}) = \mathbf{B}_{\mathbf{U}} \cdot \text{vec}(\mathbf{V})$, where $\mathbf{B}_{\mathbf{U}} = \text{circ}[\text{circ}(U_1), \dots, \text{circ}(U_N)]$, where

$$\text{circ}(a_0, \dots, a_{N-1}) \triangleq \begin{bmatrix} a_0 & a_{N-1} & \dots & a_1 \\ a_1 & a_0 & & a_2 \\ \vdots & \ddots & \ddots & \vdots \\ a_{N-1} & a_{N-2} & \dots & a_0 \end{bmatrix} \quad (28)$$

Using the above lemma, we are allowed to rewrite the DD domain relation eq. (25) into some vectorized form of

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}, \quad (29)$$

where $\mathbf{n}$ stands for some possible noise, $\mathbf{x}, \mathbf{y}$ are vectorized DD domain $\mathbf{X}, \mathbf{Y}$. And $\mathbf{H}$ here plays the role of $\mathbf{B}$ in lemma 1. By construction of $\mathbf{H}$, we know it is sparse – because each of its row sum = each of its column sum = P = number of paths, while $\mathbf{H} \in \mathbb{C}^{NM \times NM}$. Such sparsity inspired people to run message passing typed algorithms to recover $\mathbf{x}$ in eq. (29). Interested readers are referred to [6] (Sec. 5.A) for details.

Put other sophisticated algorithms aside, one may notice that eq. (29) fits in the form of MMSE. Indeed, one can easily estimate some $\hat{\mathbf{x}}$ via

$$\hat{\mathbf{x}} \triangleq (\mathbf{H}^\dagger \mathbf{H} + \lambda \mathbf{I})^{-1} \mathbf{H}^\dagger \mathbf{y}. \quad (30)$$

where λ is the effective noise power. This is in fact a fair choice, and was implemented in the MatLab code demo of OTFS in the appendix of [1]. However, the problem is the complexity. Note that MMSE's main complexity comes from inverting a $NM \times NM$ matrix, which (in its worst case) is $\mathcal{O}(N^3 M^3)$, though one may use the sparsity of $\mathbf{H}$ for some possible speed up. In contrast, the time complexity of the aforementioned message passing algorithm is way lower.

5. FURTHER DISCUSSIONS

This report gives a very quick introduction of OTFS, and a go-through of all steps of OTFS, including modulation, demodulation, channel models, channel estimation (in DD domain), and data recovery (in DD domain, as well). Because the concept of DD domain communication may be new to some readers, the scribe tried to put more emphasis on that. With this said, about the RX-side signal processing part, there are abundant works chose not to operate in the DD domain, but in frequency-time or other domains to do estimation and decoding. However, those are deemed beyond the scope of this modest report.

The scribe also focuses a lot on the foundation parts and motivation of OTFS (aka. the easier parts), and tried to avoid bothering with too complicated equations, but try to convey intuition and some understanding using both pictures and equations.

Finally, the scribe wants to remark that Saif Khan Mohammed et al published two papers [4],[7] on Zak-transform

formulated OTFS, emphasized on the theoretical and fundamental difference between Zak-OTFS and the multi-carrier OTFS (the one using SFFT/ISFFT). In our report, since we do not bother with the Heisenberg's uncertainty principle, these two constructions are (for us) equivalent. However, these two very insightful papers are very worth reading.

6. REFERENCES

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